FREEDOM INTERNATIONAL SCHOOL Bengaluru

STUDENT NAME: AFREEN TAHSILDAR		TOTAL MARKS
CLASS: XI	SUBJECT: MATHEMATICS	
ROLL NO.:	DATE: 28/8/25	

KEY ANSWERS [I TERM ASSESSMENT]

SECTION A

1) Soln:

Given: $n(A) = 200$, $n(B) = 300$, $n(A \cap B) = 100$, $n(A' \cap B') = 300$, $n(U) = ?$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$n(A \cup B) = 200 + 300 - 100 = 400$

WKT,

$n(A' \cap B') = n(A \cup B)' = 300$

Now,

$n(A \cup B)' = n(U) - n(A \cup B)$

$\therefore n(U) = 300 + 400 = 700$

Ans: b) 700

2) Soln: Here $x = a$, $y = -1$

$-1 = 2a - 3 \Rightarrow 2a = 2 \Rightarrow \boxed{a = 1}$

& Here $x = 5$ & $y = b$

$b = 2(5) - 3 \Rightarrow \boxed{b = 7}$

Ans: a) $a = 1$, $b = 7$

3) Soln: WKT,

$2 - 2x - x^2 \geq 0$

$\Rightarrow x^2 + 2x - 2 \leq 0$

$\therefore x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \dfrac{-2 \pm \sqrt{4 + 8}}{2}$

$x = \dfrac{-2 \pm \sqrt{12}}{2 \times 1} = -1 \pm \sqrt{3}$

$\therefore$ The domain is $[-1-\sqrt{3}, -1+\sqrt{3}]$

Ans: c) $[-1-\sqrt{3}, -1+\sqrt{3}]$

5) **Soln:** Since: $\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)$

$$\cos^2 48^\circ - \sin^2 12^\circ = \cos(48^\circ + 12^\circ)\cos(48^\circ - 12^\circ)$$

$$= \cos(60^\circ)\cos(36^\circ)$$

$$= \frac{1}{2} \cdot \frac{1+\sqrt{5}}{4}$$

$$\cos^2 48^\circ - \sin^2 12^\circ = \frac{1+\sqrt{5}}{8}$$

Ans: a) $\frac{1+\sqrt{5}}{8}$

4) **Soln:** a) $A \cap B'$

$\cancel{(A\cup B)\cap(A'\cup C)\cap(B\cup C) = B}$.

changed

6) **Soln:** $\cos^4 x + \sin^4 x - 6\cos^2 x \sin^2 x$

$(\cos^2 x + \sin^2 x)^2 - 2\cos^2 x \sin x - 6\cos^2 x \sin x$

$1 - 8\cos^2 x \sin^2 x$

$1 - \cancel{8}^{2}\left(\frac{\sin^2(2x)}{\cancel{4}}\right)$

$1 - 2\sin^2(2x)$

$\cos(2\cdot 2x) = \cos 4x$

$\sin^2 2x = 4\sin^2 x\cos^2 x$

$\cos 2A = 1 - 2\sin^2 A$

Ans: c) $\cos 4x$

7) **Soln:** Given: $z = \left(\frac{1}{1-i}\right)^2$

$$z = \left(\frac{1}{1-i} \times \frac{1+i}{1+i}\right)^2 = \left(\frac{1+i}{1^2-(i)^2}\right)^2 = \left(\frac{1+i}{2}\right)^2$$

$$z = \frac{1^2 + i^2 + 2i}{4} = \frac{\cancel{1} + 2i - \cancel{1}}{4} = \frac{2i}{4}$$

STUDENT NAME:		TOTAL MARKS OBTAINED
CLASS:	SUBJECT:	
ROLL NO.:	DATE:	

$z = \frac{1}{2} i \Rightarrow z = 0 + \frac{1}{2} i$

$\therefore Re(z) = 0$

Ans: a) 0

8) **Soln:** Given: $x^2 - 4x + 13 = 0$

$D = b^2 - 4ac \Rightarrow (-4)^2 - 4(1)(13) = -36 < 0$

$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm i\sqrt{36}}{2} = \frac{4 \pm 6i}{2}$

$x = \cancel{2}\left(\frac{2 \pm 3i}{\cancel{2}}\right) \Rightarrow 2 \pm 3i$

Ans: b) $2 \pm 3i$

9) **Soln:** $|x + 2| \leq 5$

$-5 \leq x + 2 \leq 5$

$-7 \leq x \leq 3$

$\therefore x \in [-7, 3]$

Ans: b) $[-7, 3]$

10) **Soln:** $\square\ \square\ \square\ \square\ \square$ (0-9)

Total no. of 5-digit numbees $= 10^5$
$= 100000$

No. of 5-digit telephone numbees with no repeated digits $= {}^{10}P_5$
$= \frac{10!}{5!}$
$= 10 \times 9 \times 8 \times 7 \times 6$
$= 30,240$

$\therefore$ No. of ~~tep~~ telephone no.'s with atleast one repeated digit $= 100000 - 30240$
$= 69760$

Ans: d) 69760

11) Soln: Word 'MOTHER'

Word starting with E : $5! = 120$ ways

Word starting with H : $5! = 120$ ways

Words starting with ME : $4! = 24$ ways

Words starting with MH : $4! = 24$ ways

$\therefore$ Total no. of words $= 240 + 48 = 288$

Words starting with MOE, MOH, MOR $= 3 \times 3!$
$= 18$

Words starting with MOTE : $1 \times 2! = 2$

$\therefore$ Total no. of words before 'MOTHER'

$288 + 18 + 2 = 308$

$\therefore$ Rank of the word 'MOTHER' $= 308 + 1$
$= 309$

Ans: c) 309

12) Soln:

12 people → 4p, 4p, 4p

$\therefore$ No. of ways $= \frac{12!}{4!\,4!\,4!\,3!}$

$= \frac{12!}{(4!)^3 3!}$

$= \frac{\cancel{34650}^{5775}}{\cancel{6}_1} = 5775$

Ans: b) 5775

13) Soln: Given $x < y$ and $b < 0$, then

$$\frac{x}{b} > \frac{y}{b}$$

Ans: a) $\frac{x}{b} > \frac{y}{b}$

STUDENT NAME:		TOTAL MARKS OBTAINED
CLASS:	SUBJECT:	
ROLL NO.:	DATE:	

14) Soln: Here $l = w + 3$

$\therefore P = 2(l+w) \leq 50$

$2(w+3+w) \leq 50$

$2w + 3 \leq 25$

$2w \leq 22$

$w \leq 11$

Ans: b) 11 cm

15) Soln: Given: $x+y+z = 1$

$\left(\frac{1}{x} - 1\right)\left(\frac{1}{y} - 1\right)\left(\frac{1}{z} - 1\right) = \frac{(1-x)(1-y)(1-z)}{xyz}$.

Now,

$\frac{(\cancel{x}+y+z-\cancel{x})(x+\cancel{y}+z-\cancel{y})(x+y+\cancel{z}-\cancel{z})}{xyz}$

$\therefore \frac{(y+z)(x+z)(x+y)}{xyz}$

WKT, AM $\geq$ GM

$y+z \geq 2\sqrt{yz}$

$x+z \geq 2\sqrt{xz}$

$x+y \geq 2\sqrt{xy}$

$\therefore \frac{(y+z)(x+z)(x+y)}{xyz} \geq \frac{8\cancel{xyz}}{\cancel{xyz}} \geq 8$

$\therefore$ Minimum value is 8.

Ans: a) 8

16) Soln: Given: $a_7 = 4096$, $a = 1$

$\therefore a_7 = ar^6$

$4096 = r^6$

$\therefore 4^6 = r^6$

$\therefore \boxed{r = 4}$

Ans: d) 4

17) Soln: WKT, $(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + \dots + {}^nC_n b^n$

$(1+x)^9 = {}^9C_0 + {}^9C_1 x + {}^9C_2 x^2 + \dots + {}^9C_9 x^9$

$(1-x)^9 = {}^9C_0 - {}^9C_1 x + {}^9C_2 x^2 - \dots - {}^9C_9 x^9$

Now,

$(1+x)^9 + (1-x)^9 = {}^9C_0 + \cancel{{}^9C_1 x} + {}^9C_2 x^2 + \dots + \cancel{{}^9C_9 x^9}$
$+ {}^9C_0 \cancel{- {}^9C_1} + {}^9C_2 x^2 + \dots \cancel{- {}^9C_9 x^9}$

$= 2[{}^9C_0 + {}^9C_2 x^2 + {}^9C_4 x^4 + {}^9C_6 x^6 + {}^9C_8 x^8]$

∴ The no. of terms is 5

Ans: b) 5

18) Soln: Given: $(3x+y)^9$

General Term: $T_{r+1} = {}^nC_r a^{n-r} b^r$

Here $r=5$, $n=9$, $a=3x$, $b=y$

∴ $T_6 = {}^9C_5 (3x)^4 (y)^5$

$T_6 = \frac{9!}{4!5!} \times 81x^4 \times y^5$

$= \frac{9 \times \cancel{8} \times 7 \times \cancel{6}}{\cancel{4} \times \cancel{3} \times \cancel{2} \times 1} \times 81x^4 y^5$

$= 10206x^4y^5$

Ans: a) $10206x^4y^5$

19) Ans: changed ~~a) Both A & R are true and R is the correct explanation of A.~~
d) A is false but R is true

20) Ans: c) A is true, but R is false.

STUDENT NAME:		TOTAL MARKS OBTAINED
CLASS:	SUBJECT:	
ROLL NO.:	DATE:	

SECTION B

21) Soln: Given: $A = \{1, 2, 3\}$, $B = \{4\}$, $C = \{5\}$

LHS: $A \times (B - C)$

$\{1, 2, 3\} \times \{4\} = \{(1,4), (2,4), (3,4)\}$

RHS: $(A \times B) - (A \times C)$

$\{(1,4), (2,4), (3,4)\} - \{(1,5), (2,5), (3,5)\}$

$\{(1,4), (2,4), (3,4)\}$

$\therefore$ LHS = RHS

Hence verified.

22) Soln: Given: ${}^{22}P_{r+1} : {}^{20}P_{r+2} = 11 : 52$

$$\frac{\frac{22!}{(21-r)!}}{\frac{20!}{(18-r)!}} = \frac{11}{52}$$

$$\frac{22!}{(21-r)!} \times \frac{(18-r)!}{20!} = \frac{11}{52}$$

$$\frac{22 \times 21 \times \cancel{20!} \times \cancel{(18-r)!}}{(21-r)(20-r)(19-r)\cancel{(18-r)!} \times \cancel{20!}} = \frac{11}{52}$$

$$\cancel{22}^{\,2} \times 21 \times 52 = \cancel{11}(21-r)(20-r)(19-r)$$

$$2 \times 21 \times 52 = (21-r)(20-r)(19-r)$$

$$2 \times \cancel{3} \times 7 \times 4 \times 13 = (21-r)(20-r)(19-r)$$

$$12 \times 13 \times 14 = (21-r)(20-r)(19-r) \text{ (By inspection)}$$

$$(21-7)(20-7)(19-7) = (21-r)(20-r)(19-r)$$

$\therefore \boxed{r = 7}$

23) Soln: Given: $a_n = 2^n + 3n$, $n \in N$

~~Let~~ S_n be the sum of terms of given sequence.

$$S_n = a_1 + a_2 + a_3 + \dots a_n$$

$$S_n = (2^1 + 3 \times 1) + (2^2 + 3 \times 2) + \dots (2^n + 3 \times n)$$

$$S_n = (2 + 2^2 + 2^3 + \dots) + 3(1 + 2 + 3 + \dots + n)$$

$$S_n = 2\left(\frac{2^n - 1}{2 - 1}\right) + 3\left(\frac{n}{2}(1+n)\right)$$

$$S_n = 2(2^n - 1) + \frac{3n}{2}(n+1)$$

G.P. $S_n = \frac{a(r^n - 1)}{r-1}$

AP $S_n = \frac{n}{2}[2a + (n-1)d]$

OR

Soln: $a_4 = \frac{1}{27}$, $a_7 = \frac{1}{729}$

$ar^3 = \frac{1}{27}$, $ar^6 = \frac{1}{729}$

Now, $\frac{\cancel{a}r^6}{\cancel{a}r^3} = \frac{1}{\cancel{729}_{27}} \times \cancel{27}$

$r^3 = \frac{1}{27}$

$$\boxed{r = \frac{1}{3}}$$

Now, $ar^3 = \frac{1}{27}$

$a\left(\frac{1}{\cancel{27}}\right) = \frac{1}{\cancel{27}} \Rightarrow \boxed{a = 1}$

$$S_n = a\left(\frac{r^0 - r^n}{1 - r}\right)$$

$$S_n = 1\left(\frac{1 - (1/3)^n}{1 - (1/3)}\right)$$

$$S_n = \frac{1 - \frac{1}{3^n}}{\frac{2}{3}}$$

$$S_n = \frac{3}{2}\left(\frac{3^n - 1}{3^n}\right)$$

STUDENT NAME: | TOTAL MARKS OBTAINED
CLASS: | SUBJECT:
ROLL NO.: | DATE:

24) **Sol^n:** Given $\frac{4}{x+1} \le 3 \le \frac{6}{x+1}$, $x \ge 0$

$\frac{4}{\cancel{x+1}}\cancel{(x+1)} \le 3(x+1) \le \frac{6}{\cancel{(x+1)}}\cancel{(x+1)}$

since $x>0$
$x+1>1$
$\therefore x+1$ is also positive

$4 \le 3(x+1) \le 6$

$4 \le 3x+3 \le 6$

$4-3 \le 3x \le 6-3$

$1 \le 3x \le 3$

$\frac{1}{3} \le \frac{\cancel{3}x}{\cancel{3}} \le \frac{\cancel{3}}{\cancel{3}}$

$\frac{1}{3} \le x \le 1$

$\therefore x \in \left[\frac{1}{3}, 1\right]$ → ½

25) **Sol^n:** The angle traced by the hour hand in 12 hours = $360°$

The angle traced by the hour hand in 3hr 30min i.e $\frac{7}{2}$ hr $= \left(\frac{\cancel{360}^{\,180\ 15}}{\cancel{12}_{\,1}} \times \frac{7}{\cancel{2}}\right)^{\circ}$

$= 105°$

The angle traced by minute hand in 60 min $= 360°$

The angle traced by the minute hand in 30 min $= \frac{\cancel{360}^{\,6}}{\cancel{60}_{\,1}} \times 30 = 180°$ ½

Hence,
The required angle between the two hands $= 180° - 105°$
$= 75°$

Also,
$75° = 75 \times \frac{\pi}{180} = \frac{5\pi}{12}$ radians → ½

OR

LHS: $15\sin\frac{5\pi}{12} + 15\cos\frac{5\pi}{12} - 20\sin^3\frac{5\pi}{12} - 20\cos^3\frac{5\pi}{12}$

$\left(15\sin\frac{5\pi}{12} - 20\sin^3\frac{5\pi}{12}\right) - \left(20\cos^3\frac{5\pi}{12} - 15\cos\frac{5\pi}{12}\right)$

$5\left(3\sin\frac{5\pi}{12} - 4\sin^3\frac{5\pi}{12}\right) - 5\left(4\cos^3\frac{5\pi}{12} - 3\cos\frac{5\pi}{12}\right)$

$5\sin\left(\cancel{3}\times\frac{5\pi}{\cancel{12}\,4}\right) - 5\cos\left(\cancel{3}\times\frac{5\pi}{\cancel{12}\,4}\right)$

$5\sin\frac{5\pi}{4} - 5\cos\left(\frac{5\pi}{4}\right)$

$5\sin\left(\pi+\frac{\pi}{4}\right) - 5\cos\left(\pi+\frac{\pi}{4}\right)$

$-5\sin\frac{\pi}{4} + 5\cos\frac{\pi}{4} = \cancel{\frac{-5}{\sqrt{2}}} + \cancel{\frac{5}{\sqrt{2}}} = 0 = RHS$

SECTION C

26) Soln: Given: $f(x) = \sqrt{\frac{x-5}{3-x}}$

Since $\frac{x-5}{3-x} \geq 0$ & $\frac{x-5}{3-x} = 0 \Rightarrow \boxed{x=5}$

And, $\frac{x-5}{3-x} > 0$

$(x-5<0$ and $3-x<0)$ or $(x-5>0$ and $3-x>0)$

$\Rightarrow (x<5$ and $3<x)$ or $(x>5$ and $3>x)$

$\therefore 3<x<5$ (Not possible)

$\therefore$ Domain of $f = 3<x\leq 5$

$= (3,5]$

STUDENT NAME: | TOTAL MARKS OBTAINED
CLASS: | SUBJECT:
ROLL NO.: | DATE:

Let $y = f(x)$, clearly $f(x) \geq 0$.

$\therefore y = \sqrt{\frac{x-5}{3-x}}$

$y^2 = \frac{x-5}{3-x}$

$3y^2 - xy^2 = x - 5 \Rightarrow x + xy^2 = 3y^2 + 5$

$x(1+y^2) = 3y^2 + 5$

$x = \frac{3y^2+5}{1+y^2}$

$\therefore$ Range of $f = [0, \infty)$ $(\because f(x) \geq 0)$

1½

27) Sol^n: INTERMEDIATE

_ 2 _ 4 _ 6 _ 8 _ 10 _ 12

i) No. of ways in which vowels occupy even places $= \frac{6!}{2!\,3!} \times \frac{6!}{2!}$

$= \frac{\cancel{6}^{3} \times 4 \times 5}{\cancel{2}} \times 6 \times 4 \times 5 \times 3$

$= 21600$

ii) No. of ways of the relative order of vowels and consonants do not alter

$= \frac{6!}{3!\,2!} \times \frac{6!}{2!}$

$= \frac{\cancel{6}^{3} \times 5 \times 4}{\cancel{2}} \times 6 \times 4 \times 5 \times 3$

$= 21600$

28) Sol^n: $(1+2\sqrt{x})^5 + (1-2\sqrt{x})^5 \rightarrow (1)$

~~(1+2~~ WKT,

$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + \dots + {}^nC_n b^n$

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$$(1+2\sqrt{x})^5 = {}^5C_0(1)^5 + {}^5C_1 2\sqrt{x} + {}^5C_2(2\sqrt{x})^2 + {}^5C_3(2\sqrt{x})^3 + {}^5C_4(2\sqrt{x})^4 + {}^5C_5(2\sqrt{x})^5$$

$$= 1 + 5(2\sqrt{x}) + 10(2\sqrt{x})^2 + 10(2\sqrt{x})^3 + \frac{5!}{4!}(2\sqrt{x})^4 + (2\sqrt{x})^5$$

$$= 1 + 10\sqrt{x} + 40x + 80x\sqrt{x} + 80x^2 + 32x^2\sqrt{x} \longrightarrow (2)$$

And

$$(1-2\sqrt{x})^5 = 1 - 10\sqrt{x} + 40x - 80x\sqrt{x} + 80x^2 - 32x^2\sqrt{x} \rightarrow (3)$$

(WKT, $(a-b)^n = {}^nC_0 a^n - {}^nC_1 a^{n-1} b + \cdots$

(From (2) & (3)) Add

$$(1+2\sqrt{x})^5 + (1-2\sqrt{x})^5 = 2 + 2(40x) + 2(80x^2)$$
$$= 2 + 80x + 160x^2$$

29) **Soln:** Given: $\frac{(1+i)^2}{2-i} = x+iy$

$$\frac{1+2i+i^2}{2-i} = x+iy$$

$$\frac{\cancel{1}+2i-\cancel{1}}{2-i} = x+iy$$

$$\frac{2i}{2-i} = x+iy$$

$$\frac{2i(2+i)}{(2-i)(2+i)} = x+iy$$

$$\frac{4i+2i^2}{4-(i)^2} = x+iy$$

$$\frac{4i+2(-1)}{4+1} = x+iy$$

STUDENT NAME: | TOTAL MARKS OBTAINED
CLASS: | SUBJECT:
ROLL NO.: | DATE:

$\frac{4i-2}{5} = x+iy$

$\therefore -\frac{2}{5} + \frac{4}{5}i = x+iy$

$\therefore \boxed{x = -\frac{2}{5}}$ and $\boxed{y = \frac{4}{5}}$

Now, 1/2

$x+y = -\frac{2}{5} + \frac{4}{5} = \frac{2}{5}$

30) a) Let the four numbers be $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$

Given: $\frac{a}{\cancel{r^3}} \times \frac{a}{\cancel{r}} \times a\cancel{r} \times a\cancel{r^3} = 4096$

$a^4 = 4096$

$a^4 = 8^4$

$\boxed{a = 8}$ — 1/2

Sum = 85

$\frac{a}{r^3} + \frac{a}{r} + ar + ar^3 = 85$

$a\left(\frac{1}{r^3} + \frac{1}{r} + r + r^3\right) = 85$

$8\left(\frac{1}{r^3} + r^3 + \frac{1}{r} + r\right) = 85$

$8\left(\frac{1}{r^3} + r^3\right) + 8\left(\frac{1}{r} + r\right) = 85$

$\cancel{8}\left[(a+b)^3 = a^3 + b^3 + 3ab(a+b)\right]$

$8\left[\left(\frac{1}{r}+r\right)^3 - 3\left(\frac{1}{r}+r\right)\right] + 8\left(\frac{1}{r}+r\right) = 85$

$8\left(\frac{1}{r}+r\right)^3 - 24\left(\frac{1}{r}+r\right) + 8\left(\frac{1}{r}+r\right) = 85$

$$8\left(\frac{1}{r}+r\right)^3-16\left(\frac{1}{r}+r\right)-85=0$$

put $\frac{1}{r}+r=x$

$$8x^3-16x-85=0$$

$$(2x-5)(4x^2+10x+17)=0$$

$2x-5=0$ $\qquad$ ($4x^2+10x+17$ has no real roots)

$$\boxed{x=\frac{5}{2}}$$

$\Rightarrow r+\frac{1}{r}=\frac{5}{2} \Rightarrow 2r^2-5r+2=0$

$\Rightarrow (r-2)(2r-1)=0$

$\Rightarrow r=2$ or $r=\frac{1}{2}$

$\therefore$ When $a=8$ and $r=2$ $\quad$ 1/2

The four numbers are

1, 4, 16, 64

When $a=8$ and $r=\frac{1}{2}$ $\quad$ 1/2

64, 16, 4, 1

b) OR Let 'a' be the first term & r be the common ratio of a G.P.

Now, $a+ar^{(n-1)}=66 \rightarrow (1)$ $\quad (r>1)$

$$ar\times ar^{(n-2)}=128$$

$$a^2r^{(n-1)}=128$$

$$ar^{(n-1)}=\frac{128}{a} \rightarrow (2)$$

from (1)

$$a+ar^{n-1}=66$$

$$a+\frac{128}{a}=66$$

$$a^2-66a+128=0$$

STUDENT NAME:
CLASS: SUBJECT:
ROLL NO.: DATE:
TOTAL MARKS OBTAINED

$a(a-2) - 64(a-2) = 0$

$(a-2)(a-64) = 0$

$\therefore \quad a = 2 \text{ or } a = 64$

putting $a = 2$ in eqn (2)

$r^{n-1} = \frac{\cancel{128}^{32}}{\cancel{4}}$

$r^{n-1} = 32 \rightarrow (3)$

putting $a = 64$ in eqn (2)

$r^{n-1} = \frac{\cancel{128}^{\cancel{2}}}{\cancel{64}\times \cancel{64}_{32}} = \frac{1}{32}$ $(\because r > 1)$

(Not considering)

$\therefore \; a = 2$ and $r^{n-1} = 32$

Now, $S_n = 126$

$a\left(\frac{r^n - 1}{r-1}\right) = 126$

$\cancel{2}\left(\frac{(r^{n-1} \times r) - 1}{r-1}\right) = \cancel{126}^{63}$

$\frac{32r - 1}{r-1} = 63 \Rightarrow 32r - 1 = 63r - 63$

$\Rightarrow \cancel{31}r = \cancel{62}^{2} \Rightarrow \boxed{r = 2}$

Now, consider eqn (3)

$r^{n-1} = 32$

$2^{n-1} = 32$

$2^{n-1} = 2^5 \Rightarrow n-1 = 5$

$\Rightarrow \boxed{n = 6}$

31) a) Soln:

LHS: $\cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) + \cos^2\left(x - \frac{\pi}{3}\right)$

$= \left(\frac{1 + \cos 2x}{2}\right) + \left(\frac{1 + \cos 2\left(x + \frac{\pi}{3}\right)}{2}\right) \left(\frac{1 + \cos 2\left(x - \frac{\pi}{3}\right)}{2}\right)$

$$\Rightarrow \frac{1}{2}\left[\left(1+\cos 2x\right)+\left(1+\cos\left(2x+\frac{2\pi}{3}\right)\right)+\left(1+\cos\left(2x-\frac{2\pi}{3}\right)\right)\right]$$

$$\Rightarrow \frac{1}{2}\left[3+\cos 2x+\cos\left(2x+\frac{2\pi}{3}\right)+\cos\left(2x-\frac{2\pi}{3}\right)\right]$$

$$\Rightarrow \frac{1}{2}\left[3+\cos 2x+2\cos 2x\cos\left(\frac{2\pi}{3}\right)\right]$$

$\cos(A+B)+\cos(A-B)=2\cos A\cos B$

$$\Rightarrow \frac{1}{2}\left[3+\cos 2x+\cancel{2}\cos 2x\left(\frac{-1}{\cancel{2}}\right)\right]$$

$$\Rightarrow \frac{1}{2}\left[3+\cancel{\cos 2x}-\cancel{\cos 2x}\right]$$

$$\Rightarrow \frac{3}{2} = RHS$$

OR

b) LHS: $\cos 2x\cos\left(\frac{x}{2}\right)-\cos 3x\cos\left(\frac{9x}{2}\right)$

WKT: $2\cos A\cos B=\cos(A+B)+\cos(A-B)$

$\cos A\cos B=\frac{1}{2}\left[\cos(A+B)+\cos(A-B)\right]$

$$\frac{1}{2}\left[\cos\left(2x+\frac{x}{2}\right)+\cos\left(2x-\frac{x}{2}\right)\right]-\frac{1}{2}\left[\cos\left(3x+\frac{9x}{2}\right)+\cos\left(3x-\frac{9x}{2}\right)\right]$$

$$\frac{1}{2}\left[\cos\frac{5x}{2}+\cos\frac{3x}{2}\right]-\frac{1}{2}\left[\cos\frac{15x}{2}+\cos\frac{3x}{2}\right]$$

STUDENT NAME:		TOTAL MARKS OBTAINED
CLASS:	SUBJECT:	
ROLL NO.:	DATE:	

$$= \frac{1}{2}\left[\cos\frac{5x}{2} + \cancel{\cos\frac{3x}{2}} - \cos\frac{15x}{2} - \cancel{\cos\frac{3x}{2}}\right]$$

$$= \frac{1}{2}\left[\cos\frac{5x}{2} - \cos\frac{15x}{2}\right]$$

$$\left\{ \text{WKT: } \cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)\right\}$$

$$= \frac{1}{2}\left[-2\sin\left(\frac{5x/2+15x/2}{2}\right)\sin\left(\frac{5x/2-15x/2}{2}\right)\right]$$

$$= \frac{1}{2}\left[-2\sin\left(\frac{\cancel{20}^{10}x/2}{\cancel{2}}\right)\sin\left(\frac{-\cancel{10}^{5}x/2}{\cancel{2}}\right)\right]$$

$$= \frac{1}{2}\left[-2\sin\left(\frac{\cancel{10}^{5}x}{\cancel{2}}\right)\sin\left(\frac{-5x}{2}\right)\right]$$

$$= \frac{1}{\cancel{2}}\left[\cancel{2}\sin 5x \sin\left(\frac{5x}{2}\right)\right]$$

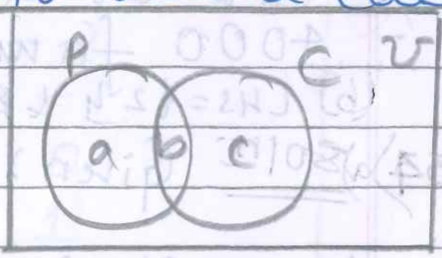
$$= \sin 5x \sin\left(\frac{5x}{2}\right) = \text{RHS}$$

SECTION D

3a) (a) Soln: Let U = all the families of the town

P = set of families who own a phone

C = set of families who own a car



Let a, b, c denotes the percentage of the families

Given

$$a+b = 25\% \rightarrow (1)$$

$b+c = 15\% \longrightarrow (2)$

$a+b+c = 100\% - 65\%$

$a+b+c = 35\% \longrightarrow (3)$

Solving (1), (2) & (3)

$a+b+c = 35\%$

$25\% + c = 35\%$

$c = 35\% - 25\%$

$\boxed{c = 10\%}$

$a+b+c = 35\%$

$a + 15\% = 35\%$

$a = 35\% - 15\%$

$\boxed{a = 20\%}$

And, $a+b = 25\% \Rightarrow \boxed{b = 5\%}$

Now let 'x' be the number of families ~~own~~ living in the town

$\therefore 5\%$ of $x = 2000$

$\frac{\cancel{5}}{100} \times x = \cancel{2000}\ 400$

$\boxed{x = 4000}$

$\therefore$ 35% of families own either a car or a phone.

$\therefore$ 4000 families live in the town.

(b) LHS = {2} & RHS = {2}

34) a) Soln: Given: $R = \{(a,b) : a, b \in Z$ and $(a-b)$ is even$\}$

i) Reflexive: Since $a \in Z \Rightarrow (a-a)$ is even

$\Rightarrow 0$ is even

$(a-a)$ is even

STUDENT NAME:		TOTAL MARKS OBTAINED
CLASS:	SUBJECT:	
ROLL NO.:	DATE:	

$\Rightarrow (a, a) \in R$
$\therefore R$ is reflexive

2) Symmetric: $\forall (a, b) \in R \Rightarrow (a-b)$ is even
$\therefore a-b$ is even
$-(a-b)$ is even
$(b-a)$ is even
$(b, a) \in R$
$\therefore R$ is symmetric

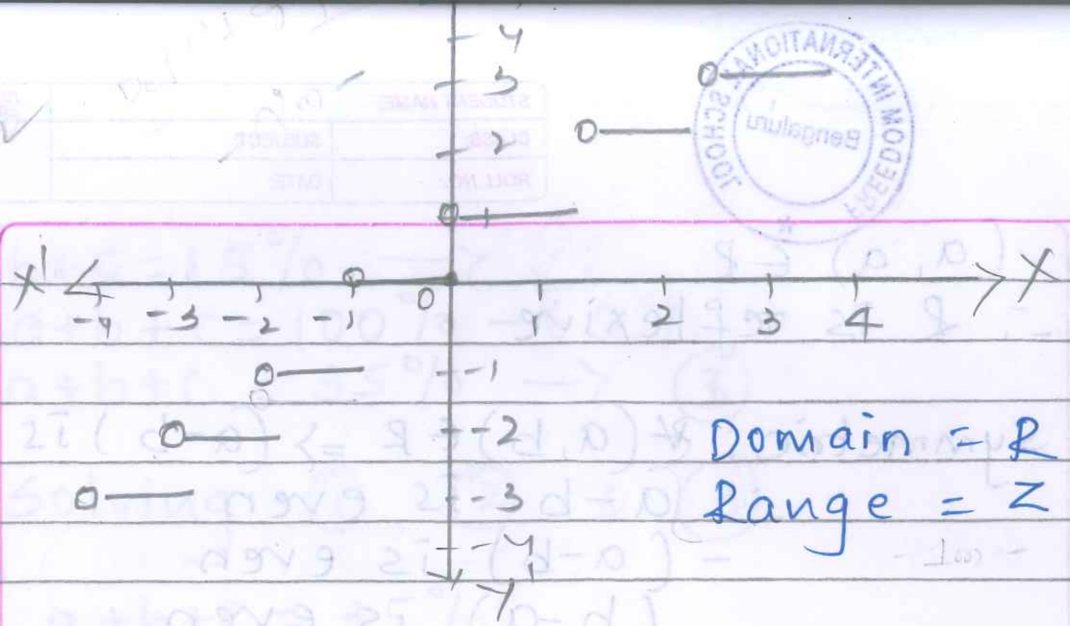
3) Transitive: $\forall (a, b) \in R \Rightarrow a-b$ is even
$(b, c) \in R \Rightarrow b-c$ is even
$\therefore (a-b) + (b-c)$ is even
$(a - \cancel{b} + \cancel{b} - c)$ is even
$(a-c)$ is even
$\therefore (a, c) \in R$
$\therefore R$ is transitive
$\therefore R$ is an equivalence relation.

OR

b) i) For any real numbee x, the smallest integee greatee than or equal to x is denoted by $\lceil x \rceil$
The function $f: R \rightarrow R$ defined as $f(x) = \lceil x \rceil$ for all $x \in R$ is called the smallest integee function.

$1\frac{1}{2}$

x	$f(x) = \lceil x \rceil$
$-3 < x \leq -2$	-2
$-2 < x \leq -1$	-1
$-1 < x \leq 0$	0
$0 < x \leq 1$	1
$1 < x \leq 2$	2
$2 < x \leq 3$	3



Domain = R
Range = Z

ii) $f(x) = \frac{kx}{x+1}$, $x \neq -1$ & $f\{f(x)\} = x$

$f(f(x)) = f\left(\frac{kx}{x+1}\right)$

$$x = \frac{k \times \frac{kx}{x+1}}{\frac{kx}{x+1} + 1} = \frac{k^2 x}{\cancel{x+1}} \times \frac{\cancel{x+1}}{kx+x+1} = \frac{k^2 \cancel{x}}{kx+x+1} = \cancel{x}$$

$\therefore k^2 \cancel{x} = kx + x + 1 \Rightarrow k^2 - kx - x - 1 = 0.$

$\therefore k^2 - kx - (x+1) = 0$

$\therefore k = \frac{x \pm \sqrt{x^2 + 4(x+1)}}{2}$

$k = \frac{x \pm (x+2)}{2} \Rightarrow k = x+1$

$\Rightarrow \boxed{k = -1}$

35) a) LHS:

$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$

$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\left(\pi-\frac{3\pi}{8}\right)\right)\left(1+\cos\left(\pi-\frac{\pi}{8}\right)\right)$

$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1-\cos\frac{3\pi}{8}\right)\left(1-\cos\frac{\pi}{8}\right)$

$\left\{\left(1+\cos\frac{\pi}{8}\right)\left(1-\cos\frac{\pi}{8}\right)\right\}\left\{\left(1+\cos\frac{3\pi}{8}\right)\left(1-\cos 3\pi\right)\right\}$

STUDENT NAME. | TOTAL MARKS OBTAINED
CLASS: | SUBJECT:
ROLL NO.: | DATE:

$$\left(1-\cos^2\frac{\pi}{8}\right)\left(1-\cos^2\frac{3\pi}{8}\right)$$

$$\sin^2\frac{\pi}{8}\sin^2\frac{3\pi}{8}$$

$\because \boxed{2\sin^2 A = 1-\cos 2A}$

$$\frac{1}{4}\left(2\sin^2\frac{\pi}{8}\right)\left(2\sin^2\frac{3\pi}{8}\right)$$

$$\frac{1}{4}\left\{\left(1-\cos\frac{\pi}{4}\right)\left(1-\cos\frac{3\pi}{4}\right)\right\}$$

$$\frac{1}{4}\left\{\left(1-\frac{1}{\sqrt{2}}\right)\left(1+\frac{1}{\sqrt{2}}\right)\right\} = \frac{1}{4}\left(1-\frac{1}{2}\right)$$

$$= \frac{1}{8} = RHS$$

OR

LHS: $\cancel{\dfrac{\sin 8x\cos x - \sin 6x\cos 3x}{\cos 2x\cos x - \sin 4x\sin 3x}}$

LHS: $\dfrac{\cos 2A\cos 3A - \cos 2A\cos 7A + \cos A\cos 10A}{\sin 4A\sin 3A - \sin 2A\sin 5A + \sin 4A\sin 7A}$

$$\frac{2\cos 3A\cos 2A - 2\cos 7A\cos 2A + 2\cos 10A\cos A}{2\sin 4A\sin 3A - 2\sin 5A\sin 2A + 2\sin 7A\sin 4A}$$

$$\frac{(\cancel{\cos 5A}+\cos A) - (\cancel{\cos 9A}+\cancel{\cos 5A}) + (\cos 11A + \cancel{\cos 9A})}{(\cos A - \cos 7A) - (\cos 3A - \cos 7A) + (\cos 3A - \cos 11A)}$$

$$\frac{\cos A + \cos 11A}{\cos A - \cancel{\cos 7A} - \cancel{\cos 3A} + \cancel{\cos 7A} + \cancel{\cos 3A} - \cos 11A}$$

$$= \frac{\cos A + \cos 11A}{\cos A - \cos 11A}$$

$$= \frac{\cancel{2}\cos\left(\frac{11A+A}{2}\right)\cos\left(\frac{11A-A}{2}\right)}{\cancel{2}\sin\left(\frac{A+11A}{2}\right)\sin\left(\frac{11A-A}{2}\right)}$$

$\frac{\cos 6A \cos 5A}{\sin 6A \sin 5A} = \cot 6A \cot 5A = RHS$

WKT,

$2\cos A \cos B = \cos(A+B) + \cos(A-B)$

$2\sin A \sin B = \cos(A+B) - \cos(A-B)$

$\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

$\cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

33) Soln: a) Given: $\frac{2z_1}{3z_2}$ is purely imaginary

$\therefore \frac{2z_1}{3z_2} = ki$ for some $k \in R$

$\frac{z_1}{z_2} = \frac{3k}{2}i$

Now, $\left|\frac{z_1 - z_2}{z_1 + z_2}\right| = \left|\frac{z_1/z_2 - 1}{z_1/z_2 + 1}\right|$

$= \left|\frac{3k/2\, i - 1}{3k/2\, i + 1}\right|$

$= \left|\frac{3ki - 2}{3ki + 2}\right|$

$= \left|\frac{-2 + 3ki}{2 + 3ki}\right|$

$= \frac{\cancel{\sqrt{4 + 9k^2}}}{\cancel{\sqrt{4 + 9k^2}}} = 1$

$\therefore \left|\frac{z_1 - z_2}{z_1 + z_2}\right| = 1$ ½

b) Given $z = 2 - 3i \Rightarrow z - 2 = -3i$

$(z-2)^2 = (-3i)^2 \Rightarrow z^2 - 4z + 4 = 9i^2$

$\Rightarrow z^2 - 4z + 13 = 0$

done in last page

STUDENT NAME:
CLASS: SUBJECT:
ROLL NO.: DATE:
TOTAL MARKS OBTAINED

~~Also, $4z^3 - 3z^2 + 169$~~
~~$4z.z^2 - 3z^2 + 169$~~
~~$4z(z^2 - 4z + 13) - 3(z^2 - 4z + 13) = 0$.~~

SECTION E

36) i) Soln: Here $a = 4$, $r = 4$, $a_6 = ?$

$a_6 = ar^5 = 4(4)^5 = 4^6 = 4096$

And, $S_5 = a\left(\frac{r^n - 1}{r - 1}\right)$

$S_5 = 4\left(\frac{4^5 - 1}{4 - 1}\right)$

$S_5 = 4\left(\frac{4^5 - 1}{3}\right) = 4\left(\frac{\cancel{1023}^{341}}{\cancel{3}}\right)$

$= 4(341) = 1364$

$S_5 = 1364$

ii) Given: $S_n > 50000$

$a\left(\frac{r^n - 1}{r - 1}\right) > 50000$

$4\left(\frac{4^n - 1}{3}\right) > 50000$

$\frac{4^n - 1}{3} > 12500$

$4^n - 1 > 37500$

$4^n > 37501$

$4^n > 4^8$

$\therefore \boxed{n = 8}$

38) i) No. of ways of arranging 3 letters out of 7 $= {}^7P_3 = \frac{7!}{4!} = 7 \times 6 \times 5 = 210$

(RNA)

No. of ways of arranging digits (RA) $= 6^4$
$= 1296$

∴ By FMC

∴ The no. of ways of different passcodes $= 210 \times 1296 = 272,160$

ii) No. of ways of arranging letters $= {}^7P_3$
$= 210$

$\underset{6}{\_\_}$ $\underset{6}{\_\_}$ $\underset{6}{\_\_}$ $\underset{3}{\_\_}$

No. of ways of arranging digits $= 6 \times 6 \times 6 \times 3$
$= 648$

∴ Total no. of passcodes can be arranged $= 210 \times 648$
$= 136,080$

iii) a) ∴ There are 3 letters, in which atleast one letter is 'E'

Remaining 2 position $= {}^6P_2 = \frac{6!}{4!} = 6 \times 5 = 30$

∴ Total no. of ways to arrange the letter E $= 3 \times 30 = 90$

$=$ (Total letter − No vowel 'E')
$({}^7P_3 - {}^6P_3)$

∴ No. of ways of arranging digits $= 6^4$
$= 1296$

∴ Total number of passcodes $= 90 \times 1296$
$= 116,640$

OR

The first letter cannot be vowel

∴ 1st letter can be filled in 5 ways

∴ 2nd & 3rd letter spots $= {}^6P_2 = 30$

∴ Total ways of letters $= 5 \times 30 = 150$

Last digit must be prime number

∴ No of ways $= 3$

STUDENT NAME. | TOTAL MARKS OBTAINED
CLASS: | SUBJECT:
ROLL NO.: | DATE:

1st digit, 2nd digit & 3rd digit has 6 ways

$\therefore$ No. of ways for digit $= 6 \times 6 \times 6 \times 3$
$= 648$

$\therefore$ Total no. of passcodes $= 150 \times 648$
$= 97200$

37) i) Given : $z = 3+4i$

$\boxed{\bar{z} = 3-4i}$

ii) $z^{-1} = \dfrac{\bar{z}}{|z|^2}$

$|z| = \sqrt{a^2+b^2}$

$z^{-1} = \dfrac{3-4i}{9+16} = \dfrac{3-4i}{25}$

$\boxed{z^{-1} = \dfrac{3}{25} - \dfrac{4}{25}i}$

iii) a) $z\bar{z} = (3+4i)(3-4i)$
$= (3)^2 - (4i)^2$
$= 9 - 16(i)^2$
$= 9+16$

$\boxed{z\bar{z} = 25.}$

And $I = Vz^{-1}$

$I = \cancel{25}\left(\dfrac{3-4i}{\cancel{25}}\right)$

$\boxed{I = 3-4i}$

(iii)(b) OR

$P = |I|^2 R$

$P = 100 \times 6$

$\boxed{P = 600}$

33) b) Soln: Given : $|z_1| = 1$

Let $z_1 = x+iy$

$z_2 = \left(\dfrac{z_1 - 1}{z_1 + 1}\right)$

$$z_2 = \frac{x+iy-1}{x+iy+1}$$

$$z_2 = \frac{(x-1)+iy}{(x+1)+iy} \times \frac{(x+1)-iy}{(x+1)-iy}$$

$$z_2 = \frac{(x^2+y^2-1)+2iy}{(x+1)^2-(iy)^2}$$

$$z_2 = \frac{(x^2+y^2-1)+2iy}{(x+1)^2+y^2}$$ $\because |z_1| = 1$

$\sqrt{x^2+y^2} = 1$

$$Re(z_2) = \frac{x^2+y^2-1}{(x+1)^2+y^2} = \frac{1-1}{(x+1)^2+y^2}$$

$$Re(z_2) = 0$$